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INFO 4220 – Networks II: Market Design
Spring 2025 – Prelim 1

Problem 1 (16 points, 4 points for each question)

1.1 The owner of three hospitals decides to use the D.A. algorithm for assigning doctors to hospitals. Based on the true preferences of hospitals and doctors they run the D.A. algorithm with hospitals and doctors proposing and are trying to decide which version they should implement. Each hospital has a capacity for two doctors, and there are five applicants. They get the following matches:

Hospital Proposing: $\mu(h_1) = (d_4, d_5), \mu(h_2) = (d_3, \emptyset), \mu(h_3) = (d_1, d_2)$

Doctors proposing: $\mu(h_1) = (d_1, d_2), \mu(h_2) = (d_3, \emptyset), \mu(h_3) = (d_4, d_5)$

Are both results stable matches? Explain why yes/no.

If the owner of the hospitals cares about hospitals getting their most preferred doctors, which version of the algorithm should they use?

Answer:

Yes, both results are stable. (1pt) Because the matching produced by the DA algorithm are stable. (1pt)

The owner should prefer the version with the hospital proposing, because it gives the hospital optimal matching, which is the most preferred stable matchings for hospitals. (2pt)

1.2 In the upcoming market for quantum computing chips it is estimated that demand and supply for chips are given by the following equations:

$Q_s = 4P - 4$ and $Q_d = 20 - 2P$

The quantum chip manufacturers are figuring out production and will only be able to produce 9 chips. Calculate the price chips will be sold, the consumer surplus, and producer surplus. Show these quantities in a simple graph (no need to make it to scale)

Answer:

When there is no quantity quota,

$$Q_s = 4P - 4 = Q_d = 20 - 2P \Rightarrow P = 4, Q = 12$$

Given the quantity cap 9, the price to be sold will be given by

$Q_d = 20 - 2P = 9 \Rightarrow P = 5.5 = \frac{11}{2}$ (0.5pt if shown in the graph, same for below)

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Consumer surplus:

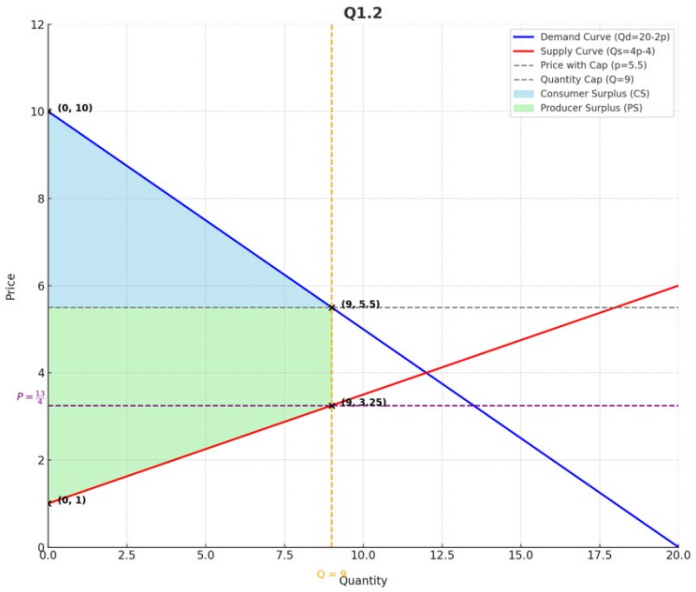
$CS = \frac{1}{2} * 9 * (10 - 5.5) = 20.25 = \frac{81}{4}$ (0.5pt)

Producer surplus:

$Q_s = 9 = 4P - 4 \Rightarrow P = \frac{13}{4} = 3.25$ (0.5pt)

$PS = \frac{1}{2} * 9 * (5.5 - 3.25 + 5.5 - 1) = 30.375 = \frac{243}{8}$ (0.5pt)

Graph: 2pt for the right area of CS and PS.



Partial credit (<=2pt) will be given if answering the question without the quantity cap at 9.

1.3 Alice and Bob are deciding what movie to watch together. Their payoffs are given in the matrix below.

	Bob			
Alice		Action	Comedy	Horror
	Action	(50, 40)	(0, 70)	(10, 0)
	Comedy	(10, 0)	(30, 30)	(10, 10)
	Horror	(0, 0)	(0, 30)	(10, 30)

Find the Nash Equilibrium(a) of this game. Are any of the N.E. Pareto efficient? What do you expect will be the outcome of this game? Justify your answer.

Answer:

There are 2 N.E. of the game: (Comedy, Comedy), (Horror, Horror). (1pt for each, -0.5 if listing other outcome)

Neither is Pareto efficient (0.5pt), as (Action, Action) is pareto superior to both. (0.5pt)

I would expect the game to end in (Comedy, Comedy). (1pt)

1.4 4 students (Alice, Bob, Carol, Denis) are to be assigned rooms (A-D) in a dorm. Their preferences are shown in the table below. Use Serial Dictatorship to find the assignment. Follow the order of dictators: Carol, Alice, Denis, Bob. Is this assignment efficient? Explain why? Would your answer change if a different order of dictators is used?

Alice	**Bob**	**Carol**	**Denis**
B	B	B	C
A	C	D	D
C	D	A	A
D	A	C	B

Answer:

Carol -> B, Alice -> A, Denis -> C, Bob -> D. (1pt)

The assignment is efficient. (1pt) Because no one can be better off without making some else worse. (1pt)

The order doesn't matter. Serial dictatorship always generates efficient outcomes. (1pt)

Alternative for "Would your answer change if a different order of dictators is used?":

The result for the matching might change if the order is changed. However, the matching will still be efficient. (1pt)

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Problem 2 (17 points)

In a one-to-one matching problem, there are 4 musicians and 4 singers who are trying to form pairs for Slope Day. For **this year**, they have the following preferences.

Musician				**Singer**			
M1	**M2**	**M3**	**M4**	**S1**	**S2**	**S3**	**S4**
S2	S1	S2	S4	M2	M2	M4	M1
S1	S2	S1	S3	M4	M3	M1	M2
S4	S4	S3	S1	M3	M1	M3	M3
S3	S3	S4	S2	M1	M4	M2	M4

a) Last year, M1's preferences list was S3, S2, S1, S4. The matching for last year was M1 -> S3, M2 -> S1, M3 -> S2, M4 -> S4. Is this matching stable if we use this year's preferences? Explain why or why not. (4 pts)

b) Use the deferred acceptance algorithm with musician proposing to find a matching for this year. Show all your steps. (Note: for part (b) – (e), use M1's preference is as in the table above: S2, S1, S4, S3.) (4 pts)

c) Is the algorithm in (b) strategyproof for M1? If yes, explain why. If not, provide an example when M1 would benefit from misrepresenting his preference. (3 pts)

d) Is there any stable matching that singers will prefer to the one you found in part (b). (3 pts)

e) Will singers have any incentives to misrepresent their preferences in the algorithm in part (b) if we assume that musicians will not misrepresent their preferences? What does this mean for the strategyproofness of the algorithm in part (b) for singers? (3 pts)

Answer:

a) The matching is unstable. (2pt)
We have blocking pairs (M1, S4). (2pt)

b) Step 1: M1 -> S2, M2 -> S1, M3 -> S2, M4 -> S4. S2 rejects M1.
Step 2: M1 -> S1. S1 rejects M1.
Step 3: M1 -> S4. S4 accepts M1 and rejects M4.
Step 4: M4 -> S3. No further rejection.
(0.5pt for each step)
Final matching: M1 -> S4, M2 -> S1, M3 -> S2, M4 -> S3. (2pt for final matching)

c) Yes, algorithm in b is strategyproof for M1. (2pt)
Because DA with musician proposing is strategyproof for the proposing side. (1pt)

d) No. There is only 1 stable matching here. The matching from DA with singers proposing is the same as the one from b.
Grading rubric: Realizes that DA with singers proposing produces the singer optimal match (1.5pt)
The matching from DA with singers proposing is the same as the one from b. (1.5pt)

e) No (1pt), because singers already get the singer-optimal matching by being truthful. (1pt)
This does not mean anything for strategyproofness. (1pt)

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Problem 3 (17 points)

Three hospitals are deciding which doctors to hire from a pool of seven doctors. The hospitals have the following capacity and preferences:

h_1 (capacity = 2)	h_2 (capacity = 2)	h_3 (capacity = 3)
d_1	d_2	d_3
d_3	d_4	d_5
d_6	d_6	d_6
d_7	d_7	d_7
d_2	d_1	d_1
d_4	d_3	d_2
d_5	d_5	d_4

The doctors have the following preferences:

d_1	d_2	d_3	d_4	d_5	d_6 & d_7 (couple)
h_1	h_2	h_3	h_2	h_3	h_1 & h_1
h_2	h_1	h_1	h_3	h_2	h_2 & h_2
h_3	h_3	h_2	h_1	h_1	h_3 & h_3

a) Use D.A. with doctors proposing to find a matching. Note that doctors in the couple make offers together and if one doctor in the couple is rejected, the other doctor withdraws their offer and the couple makes a new offer as a couple in the next step. Is the result stable? (4 pts)
b) Assume hospitals will prefer to fill up their capacity rather than stick to their preferences. To do so, when analyzing couples, they will choose the scenario of full capacity rather than having empty spots and a preferred doctor. (5 pts)
c) For each hospital, and for each doctor, explain if they prefer the matching you found in part (a) or part (b) (4 pts)
d) Is the matching you found in part (b) stable? (4 pts)

Answer:

a) Step 1: d1 -> h1, d2 -> h2, d3-> h3, d4 -> h2, d5 -> h3, {d6, d7} -> h1
h1 rejects {d6, d7} (1pt)
Step 2: {d6, d7} -> h2. H3 rejects {d6, d7}. (0.5pt)
Step 3: {d6, d7} -> h3. H3 rejects {d6, d7}. (0.5pt)
Final matching: h1 -> {d1}, h2 -> {d2, d4}, h3 -> {d3, d5}. d6 and d7 remain unmatched. (1pt)
The result is stable. Because we need to consider couples as pair here, and there is no individual irrational, no blocking pair, and no wastefulness.(1pt)

b) Step 1: d1 -> h1, d2 -> h2, d3-> h3, d4 -> h2, d5 -> h3, {d6, d7} -> h1
h1 prefers {d6, d7} than {d1, Ø}, so h1 rejects d1. (2pt)
Step 2: d1 -> h2. H2 rejects d1. (1pt)
Step 3: d1 -> h3. H3 accepts d1 and no more rejections. (1pt)
Final matching: h1 -> {d6, d7}, h2 -> {d2, d4}, h3 -> {d1, d3, d5}. (1pt)

c) h2 will be indifferent because they have the same matching. d2, d3, d4, d5 will be indifferent because they have the same matching. (1pt)

H3 will prefer result in (b), because they fulfill their capacity. D1 prefers (a), which matches to more preferred hospital. D6 and d7 prefers (b), because they could match to their preferred hospital than unmatched. (1pt)

For h1, we need to compare {d6, d7}, {d1, Ø}. We can not compare those responsive preferences, so we don't know if h1 prefers (a) or (b). (2pt)
(Partial credit for h1 if only discussing the case based on assumption from b: 1pt, If we consider they keep the assumption from b, h1 would prefer b.)

d) Yes, it is stable. (2pt)
Under the original preference, we can not say that {d1, h1} form a blocking pair, because we don't know if h1 prefers {d1, Ø} to {d6, d7}. (2pt)

(Alternative answer for full credits: it is stable. (d1, h1) is not a blocking pair under the assumption from b.)